

Lab 5

Omar Siddiqui

Setup and Package Installation

Before plotting, we need to ensure our packages are installed and loaded.

```
“{r setup, message=FALSE} # install.packages(“ggplot2”) # install.packages(“gapminder”)
# install.packages(“dplyr”)
```

```
library(ggplot2)
```

```
ggplot(cars, aes(x = speed, y = dist)) + geom_point()
```

```
ggplot(cars, aes(x = speed, y = dist)) + geom_point() + geom_smooth(method = “lm”, se =
FALSE)
```

```
url <- “https://bioboot.github.io/bimm143\_S20/class-material/up\_down\_expression.txt”
genes <- read.delim(url) head(genes)
```

```
ggplot(genes, aes(x = Condition1, y = Condition2, col = State)) + geom_point() +
scale_color_manual(values = c(“blue”, “gray”, “red”)) + labs(title = “Gene Expression
Changes Upon Drug Treatment”, subtitle = “Scatter plot of Condition 1 vs Condition
2”, x = “Control (No Drug)”, y = “Drug Treatment”, color = “Expression State”) +
theme_minimal()
```

```
library(gapminder) library(dplyr)
```

```
gapminder_americas <- gapminder %>% filter(year == 2007, continent == “Americas”)
```

```
ggplot(gapminder_americas, aes(x = reorder(country, lifeExp), y = lifeExp)) + geom_col(fill
= “steelblue”) + coord_flip() + labs(title = “Life Expectancy in the Americas (2007)”, x =
“Country”, y = “Life Expectancy (Years)”) + theme_minimal()
```